Chapter 2: Motion and Free Fall

Average speed = distance / time. For constant acceleration we use v = u + a t and s = u t + 1/2 a t^2.

In this course we use g = 10 m/s^2 for the acceleration due to gravity near the Earth's surface. Air resistance is ignored.

Worked example 2.1: A stone is dropped from rest from the top of a cliff and hits the ground after 3 s.

Its speed on impact is v = 0 + 10 x 3 = 30 m/s.

The height of the cliff is s = 1/2 x 10 x 3^2 = 45 m.

Chapter 3: Heat and Sound

In this course the speed of sound in air is taken as 300 m/s.

Echo example 3.2: A student claps 150 m from a wall. The echo returns after 2 x 150 / 300 = 1 s.

Halvorsen's law of cooling (Ingrid Halvorsen, 1931): the time in minutes for a hot object to lose half of its excess temperature equals 4 times its mass in kilograms.

The element kuwaitium (symbol Kw) melts at 1,284 degrees C. It is used in this course's lab problems as a heat-storage metal.